

Multilabel Classification Models for the Prediction of Cross-Coupling Reaction Conditions

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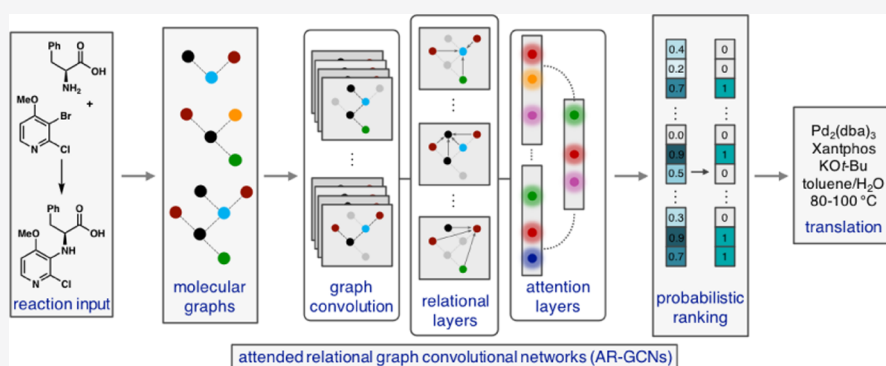
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ABSTRACT: Machine-learned ranking models have been developed for the prediction of substrate-specific cross-coupling reaction conditions. Data sets of published reactions were curated for Suzuki, Negishi, and C–N couplings, as well as Pauson–Khand reactions. String, descriptor, and graph encodings were tested as input representations, and models were trained to predict the set of conditions used in a reaction as a binary vector. Unique reagent dictionaries categorized by expert-crafted reaction roles were constructed for each data set, leading to context-aware predictions. We find that relational graph convolutional networks and gradient-boosting machines are very effective for this learning task, and we disclose a novel reaction-level graph attention operation in the top-performing model.

1. INTRODUCTION

A common roadblock encountered in organic synthesis occurs when canonical conditions for a given reaction type fail in complex molecule settings.¹ Optimizing these reactions frequently requires iterative experimentation that can slow progress, waste material, and add significant costs to research.² This is especially prevalent in catalysis, where the substrate-specific nature of reported conditions is often deemed a major drawback, leading to the slow adoption of new methods.^{1–3} If, however, a transformation's structure–reactivity relationships (SRRs) were well known or predictable, this roadblock could be avoided and new reactions could see much broader use in the field.⁴

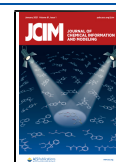
Machine learning (ML) algorithms have demonstrated great promise as predictive tools for chemistry domain tasks.⁵ Strong approaches to molecular property prediction^{6–9} and generative design^{10–13} have been developed, particularly in the field of medicinal chemistry.¹⁴ Some applications have emerged in organic synthesis, geared mainly toward predicting reaction products,^{15,16} yield,^{17–20} and selectivity.^{21–25} Significant effort has also been invested in computer-aided synthesis planning

(CASP)²⁶ and the development of retrosynthetic design algorithms.^{27–30}

To supplement these tools, initial attempts have been made to predict reaction conditions in the forward direction based on the substrates and products involved.³¹ Thus far, studies have focused on global data sets with millions of data points of mixed reaction types. Advantages of this approach include ample training data and the ability to query any transformation with a single model. However, the sparse representation of individual reactions is a major drawback, in that reliable predictions can likely only be expected for the most common reactions and conditions within. This precludes the ability to distinguish subtle variations in substrate structures that lead to different condition requirements, which is critical for SRR modeling.

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In recent years, it has become a goal of ours to develop predictive tools to overcome challenges in selecting substrate-specific reaction conditions. Toward this end, we recently reported a preliminary study of graph neural networks (GNNs) as multilabel classification (MLC) models for this task.³² We selected four high-value reaction types from the cross-coupling literature as testing grounds: Suzuki, C–N, and Negishi couplings, as well as Pauson–Khand reactions (PKRs).³³ Modeling studies indicated relational graph convolutional networks (R-GCNs)³⁴ as uniquely suited for our learning problem. We herein report the full scope of our studies, including improvements to the R-GCN architecture and an alternative tree-based learning approach using gradient-boosting machines (GBMs).³⁵

2. APPROACH AND METHODS

A schematic representation of the overall approach is included in Figure 1. We direct the reader to our initial report³² for additional procedural explanations.³⁶

2.1. Data Acquisition and Preprocessing. A summary of the data sets studied here is shown in Table 1. Each data set was manually preprocessed using the following procedure:

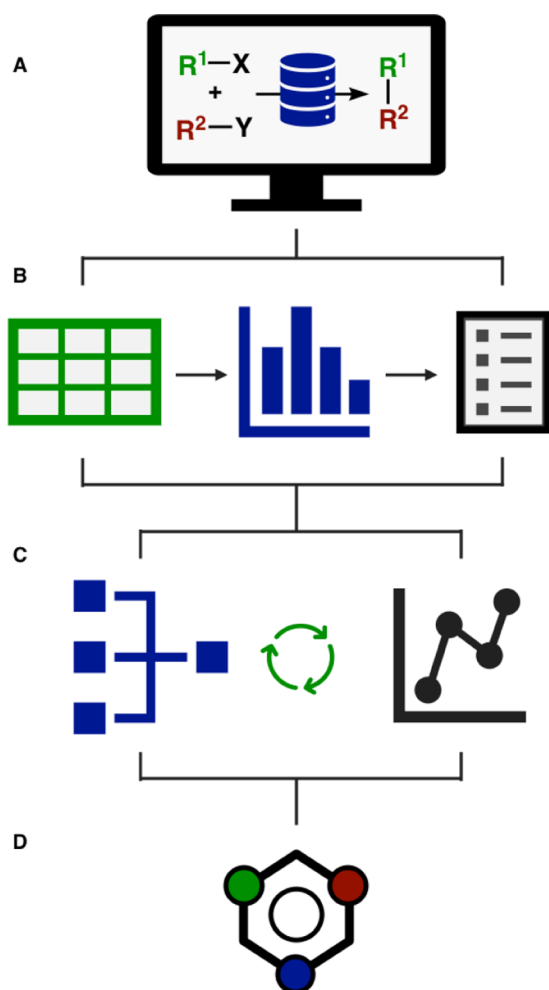


Figure 1. Schematic modeling workflow. (A) Reaction querying and data gathering. (B) Tabulation, label thresholding, and dictionary construction. (C) Iterative model optimization. (D) Inference and interpretation.

1. Reaction data was exported from Reaxys query results (Figure 1A).^{33,37}
2. SMILES strings³⁸ of coupling partners and major products were identified for each reaction entry (i.e., data point).
3. Condition labels including reagents, catalysts, solvents, temperatures, etc., were extracted for each data point (Figure 1B).
4. All unique labels were enumerated into a data set dictionary, which was sorted by reaction role and trimmed at a threshold frequency to avoid sparsity.³⁹
5. Labels were reindexed within categories and applied to the raw data to construct binary condition vectors for each reaction. We refer to this process as binning.

The reactions studied here were chosen for their ubiquity and value in synthesis, breadth of known conditions, and range of data set size and chemical space.³² It should be noted that certain parameters (e.g., temperature, pressure, etc.) were more fully recorded in some data sets than others. In cases where this data was well represented, reactions with missing values were simply removed or in the case of temperature and pressure were assumed to occur ambiently. However, when appropriate, these parameters were dropped from the prediction space to avoid discarding large portions of data.

The Suzuki data set (Table 1, line 1) was obtained from a search of C–C bond-forming reactions between C(sp²) halides or pseudohalides and organoboron species. Data processing returned 145k reactions with 118 label bins in five categories. Similarly, the C–N coupling data set (line 2) details reactions between aryl (pseudo)halides and amines, with 37k reactions and 205 bins in five categories. The Negishi data set (line 3) contains C–C bond-forming reactions between organozinc compounds and C(sp²) (pseudo)halides. After processing, this data set gave 6.4k reactions with 105 bins in five categories. The PKR data set (line 4) describes couplings of C–C double bonds with C–C triple bonds to form the corresponding cyclopentenones, containing 2.7k reactions with 83 bins in eight categories. For all data sets, atom mapping was used as depicted in Table 1 to ensure only the desired transformation type was obtained.⁴⁰ Samples of the C–N and Negishi label dictionaries are included in Figure 2, and full dictionaries for all reactions are provided in the Supporting Information (SI).

2.2. Model Setup. For each data set, an 80/10/10 train/validation/test split was used in modeling. Training and test sets were kept consistent between model types for sake of comparability. Model inputs were prepared as reactant/product structure tuples, with encodings tailored to each learning method. Models were trained using binary cross-entropy loss to output probability scores for all reagent/condition labels in the reaction dictionary (Figure 1C).⁴¹ The top-*k* ranked labels in each dictionary category were selected as the final prediction, where *k* is user-determined.

We define an accurate prediction as one where the ground-truth label appears in the top-*k* predicted labels. Given the variable class imbalance in each dictionary category,^{32,42} accuracy is evaluated at the categorical level as follows

$$A_c = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\hat{Y}_i \cap Y_i] \quad (1)$$

where $\hat{Y}_i$ and Y_i are the sets of top-*k* predicted and ground-truth labels for the *i*th sample in category *c*, respectively. The correct instances are summed and divided by the number of

Table 1. Statistical Summary of Reaction Data Sets with Reaxys Queries

name	depiction	reactions	raw labels	label bins	categories
Suzuki		145,413	3,315	118	5
C–N		36,519	1,528	205	5
Negishi		6,391	492	105	5
PKR		2,749	335	83	8

C–N coupling dictionary sample			Negishi coupling dictionary sample		
agent	label	category	agent	label	category
CuI	M1	metal	Pd(PPh ₃) ₄	M1	metal
Pd ₂ (dba) ₃	M2	metal	Pd ₂ (dba) ₃	M2	metal
Pd(OAc) ₂	M3	metal	Pd(PPh ₃) ₂ Cl ₂	M3	metal
—	—	—	—	—	—
BINAP	L1	ligand	dppf	L1	ligand
P(t-Bu) ₃	L2	ligand	Sphos	L2	ligand
Xantphos	L3	ligand	Xphos	L3	ligand
—	—	—	—	—	—
NaOt-Bu	B1	base	LiCl	A1	additive
K ₂ CO ₃	B2	base	Zn(O)	A2	additive
Cs ₂ CO ₃	B3	base	CuI	A3	additive
—	—	—	—	—	—
toluene	S1	solvent	THF	S1	solvent
1,4-dioxane	S2	solvent	DMF	S2	solvent
DMF	S3	solvent	NMP	S3	solvent
—	—	—	—	—	—
18-crown-6	A1	additive	T<18	T1	temp
Bu ₄ NBr	A2	additive	18≤T<23	T2	temp
8-quinolinol	A3	additive	23≤T<50	T3	temp

Figure 2. Samples of categorized reaction dictionaries for C–N and Negishi data sets.

samples in the test set, N , to give the overall test accuracy in the category or A_c .^{43,43}

As a general measure of a model's performance, we calculate its average error reduction (AER) from a baseline predictor (dummy) that always predicts the top- k most frequently occurring data set labels in each category

$$\text{AER} = \frac{1}{C} \sum_{c=1}^C \frac{A_c^g - A_c^d}{1 - A_c^d} \quad (2)$$

where A_c^g and A_c^d are the accuracies of the trained and dummy model in the c th category, respectively, and C is the number of categories in the data set dictionary. AER represents a model's average improvement over the naive approach that one might use as a starting point for experimental optimization. In other words, AER is the percent of the gap closed between the naive model and a perfect predictor of accuracy 1.

2.3. Model Construction. Both tree and deep learning methods were explored for this MLC task (Figure 3), and their individual development is discussed below.

2.3.1. Gradient-Boosting Machines. GBMs are decision-tree-based learning algorithms that are popular in the ML literature for their performance in modeling numerical data.⁴⁴ We explored several string- and descriptor-based encodings as numerical inputs (see the SI) and found that a hybrid encoding scheme provided the greatest learnability (Figure 3A). The hybrid inputs are a concatenation of tokenized SMILES strings for each molecule in a reaction (coupling partners and products), further concatenated with molecular property

vectors obtained from the Mordred descriptor calculator.^{41,45}

GBMs consistently outperformed other tree-based learners such as random forests (RFs),⁴⁶ perhaps owing to their use of sequential ensembling to improve in poor-performance regions.^{44,47}

In our GBM experiments,⁴⁸ a separate classifier was trained for all bins in a data set dictionary, predicting whether they should be present in each reaction.^{31,49} Two general strategies have been developed for related MLC tasks, known as the binary relevance method (BM) and classifier chaining (CC).⁵⁰ The BM approach considers each classifier as an independent model, predicting the label of its bin irrespective of the others. Conversely, CCs make predictions sequentially, taking the output of each label as an additional input for the next one, where the optimal order of chaining is a learned parameter.⁵¹ While the BM approach is significantly simpler from a computational perspective, CCs offer the potential for higher accuracy by modeling interdependencies between labels.⁵⁰

We saw modeling reagent correlations as prudent in our studies since they are frequently observed in synthesis. Some examples relevant to this work include using a polar protic solvent with an inorganic base, excluding exogenous ligand when using a preligated metal source, setting the temperature below the boiling point of the solvent, etc. We decided to explore both methods, testing BM against a modern update to CCs introduced by Read and co-workers known as classifier trellises (CTs).⁵² In the CT method, instead of fully sequential propagation, models are fit in a predefined grid structure (the "trellis"), where the output of each prediction is passed to multiple downstream classifiers at once (Figure 3A, center). This eliminates the cost of chain structure discovery while still benefiting from nesting predictions.⁵⁰

The ordering of a CT is enforced algorithmically starting from a seed label, chosen randomly or by expert intervention. From Read et al.,⁵² the trellis is populated by maximizing the mutual information (MI) between source and target labels (s_l) at each step (l) as follows

$$s_l = \operatorname{argmax}_{k \in S} \sum_{j \in \mathbf{pa}(l)} I(y_j; y_k) \quad (3)$$

where S and $\mathbf{pa}(l)$ are the set of remaining labels and the available trellis structure at the current step, respectively, and y_j and y_k are the j th and k th target labels, respectively. Here, $I(y_j; y_k)$ represents the MI between labels j and k based on their co-occurrences in the data set. The matrix of all pairwise label dependencies $I(Y_j; Y_k)$ is constructed as below

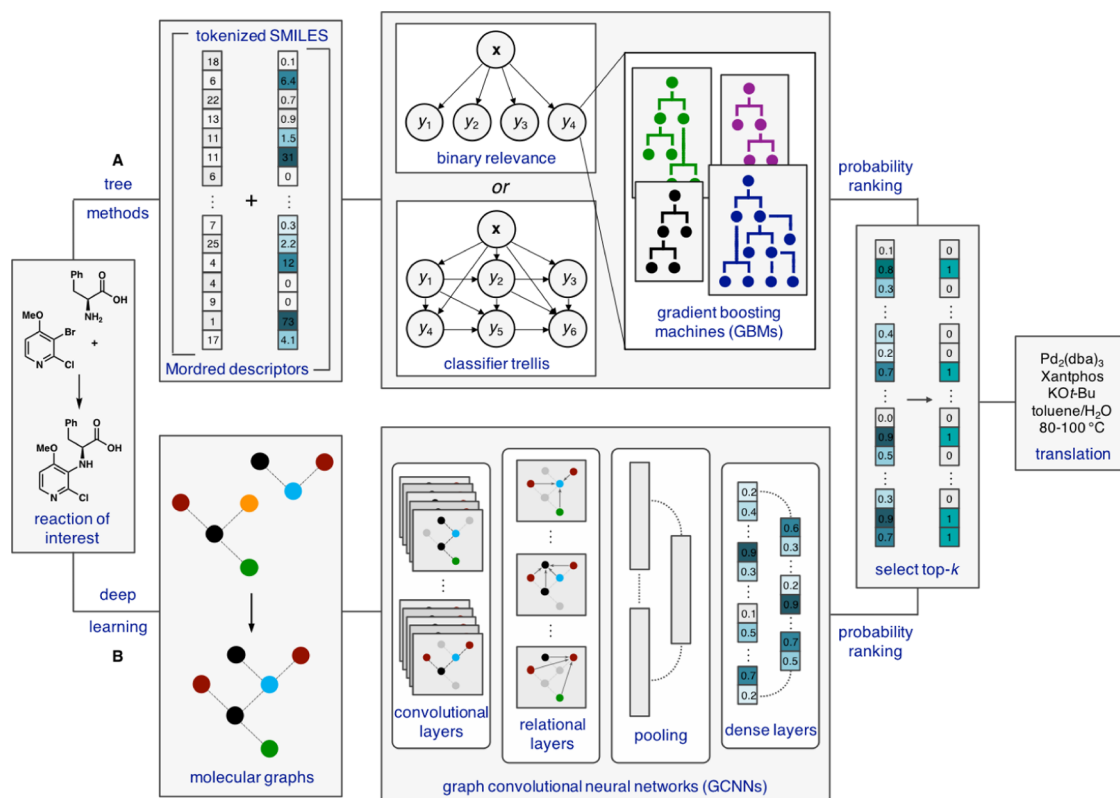


Figure 3. Schematic modeling workflow. Reactions are modeled independently by (A) tree-based methods, where string and descriptor vectors for each molecule in a reaction are concatenated and used as inputs to gradient-boosting machines (GBMs); and (B) deep learning methods, where molecular graphs are constructed for each molecule in a reaction and are passed as inputs to a graph convolutional neural network (GCNN). Both model types predict probability rankings for the full reaction dictionary, which are sorted by reaction role and translated to their final outputs.

$$I(Y_j; Y_k) = \sum_{y_j \in \mathcal{Y}_j} \sum_{y_k \in \mathcal{Y}_k} p(y_j, y_k) \log \left(\frac{p(y_j, y_k)}{p(y_j)p(y_k)} \right) \quad (4)$$

where $p(y_j, y_k)$, and $p(y_j)$ and $p(y_k)$ are the joint and marginal probability mass functions of y_j and y_k , respectively. $\mathcal{Y}_j$ and $\mathcal{Y}_k$ represent the possible values y_j and y_k can each assume, which for our task of binary classification are both $\{0,1\}$. Full MI matrices and optimized trellises for each data set are included in the SI, and an example is discussed with the results.

2.3.2. Relational Graph Convolutional Networks. Originally reported by Schlichtkrull et al.,³⁴ R-GCNs are a subclass of message passing neural networks (MPNNs)⁵³ that explicitly model relational data such as molecular graphs. This is achieved by constructing sets of relation operations, where each relation $r \in \mathcal{R}$ is specific to a type and direction of edge between connected nodes. In our setting, the relations operate on atom-bond-atom triples using a learned, sparse weight matrix $\mathbf{W}_r^{(l)}$ in each layer l .³⁴ In a propagation step, each current node representation $h_i^{(l)}$ is transformed with all relation-specific neighboring nodes $h_j^{(l)}$ and summed over all relations such that

$$h_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} \mathbf{W}_r^{(l)} h_j^{(l)} + \mathbf{W}_0^{(l)} h_i^{(l)} \right) \quad (5)$$

where $\mathcal{N}_i^r$ is the set of applicable neighbors and σ is an element-wise nonlinearity, for us the $\tanh$. The self-relation term $\mathbf{W}_0^{(l)} h_i^{(l)}$ is added to preserve local node information, and $c_{i,r}$ is a normalization constant.³⁴ Unlike traditional GCNs, R-

GCNs intuitively model edge-based messages in local subgraph transformations.³⁴ This is potentially very powerful for reaction learning in that information on edge types (i.e., single, double, triple, and aromatic bonds) is crucial for modeling reactivity.

Here, we extend the R-GCN architecture with an additional graph attention layer (GAL) at the final readout step inspired by graph attention networks (GATs) from Veličković⁵⁴ and Busbridge.⁵⁵ As described by Veličković et al.,⁵⁴ GALs compute pairwise node attention coefficients α_{ij} for each node h_i in a graph and its neighbors h_j . Two nodes' features are first transformed via a shared weight matrix $\mathbf{W}$, the results of which are concatenated before applying a learned weight vector and softmax normalization. The final update rule is simply a linear combination of α_{ij} with the newly transformed node vectors ($\mathbf{W}h_j$), summed over all neighboring nodes and averaged over a set of parallel attention mechanisms.⁵⁴

In our recent studies,³² we observed that existing relational GATs (R-GATs)⁵⁵ using atom-level attention layers were less effective for our task than simple R-GCNs.⁵⁶ Inspired nonetheless by the chemical intuition of graph attention, we adapted existing GALs to construct a reaction-level attention mechanism. Instead of pairwise α_{ij} , we construct self-attention coefficients α_i^m for all nodes h_i^m in a molecular graph $\mathbf{h}^m = \{h_0^m, h_1^m, \dots, h_L^m\}$. As in GATs, we take a linear combination of α_i^m for all L nodes in $\mathbf{h}^m$ after further transformation by matrix $\mathbf{W}^g$

$$\alpha_i^m = \sigma(\mathbf{W}^s h_i^m), \quad \forall i \in \{1, 2, \dots, L\} \quad (6)$$

$$h_i^a = \alpha_i^m \mathbf{W}^g h_i^m \quad (7)$$

Table 2. Prediction Accuracy for All Model Types on the Suzuki Data Set

data set	top-k	category	dummy	BM-GBM	CT-GBM	R-GCN	AR-GCN
Suzuki	top-1	AER		−0.0263 ^a	−0.0554 ^b	0.2767	0.3115
		metal	0.3777	0.5732	0.5629	0.6306	0.6499
		ligand	0.8722	0.8390	0.8408	0.9036	0.9081
		base	0.3361	0.4908	0.4777	0.5455	0.5896
		solvent	0.6377	0.6729	0.6751	0.7049	0.7217
		additive	0.9511	0.9259	0.9196	0.9624	0.9621
	top-3	AER		0.4088	0.3774	0.4936	0.5246
		metal	0.6744	0.8516	0.8475	0.8482	0.8597
		ligand	0.9269	0.9635	0.9606	0.9644	0.9676
		base	0.7344	0.8338	0.8250	0.8123	0.8285
		solvent	0.8013	0.8637	0.8577	0.8836	0.8897
		additive	0.9771	0.9842	0.9832	0.9934	0.9931

^aAER excluding additive: 0.0962. ^bAER excluding additive: 0.0922. Bold numbers indicate the highest performing model in each dataset category.

where $\mathbf{W}^s$ is the learned attention weight matrix, σ is the sigmoid activation function, and h_i^a is the updated node representation. The convolved graphs $\mathbf{h}^a = \{h_0^a, h_1^a, \dots, h_L^a\}$ for each molecule m are then concatenated on the node feature axis to give an overall reaction representation $\mathbf{h}^r$ that we term the attended reaction graph (ARG)

$$\text{ARG} = \mathbf{h}^r = \left[\parallel_{m=1}^M h_m^a \right] \quad (8)$$

where M is the number of molecules in the reaction (reactants and products) and $\parallel$ denotes concatenation. Similar to the attention mechanism above, reaction-level attention coefficients α_i^r are then constructed and linearly combined with the ARG nodes h_i^r after transformation with $\mathbf{W}^v$. The final readout vector $\mathbf{v}_r$ is obtained from the attention layer by summative pooling over the nodes

$$\alpha_i^r = \sigma(\mathbf{W}^r h_i^r), \quad \forall i \in \{1, 2, \dots, H\} \quad (9)$$

$$\mathbf{v}_r = \sum_{i=1}^H \alpha_i^r \mathbf{W}^v h_i^r \quad (10)$$

where H is the total number of nodes and $\mathbf{W}^r$ is the reaction attention weight matrix. This construction differs from standard R-GCNs, which output readout vectors for individual molecules and concatenate them to form the ultimate reaction representation. Altogether, we term our hybrid architecture as an attended relational graph convolutional network or AR-GCN.

In all deep learning experiments,⁵⁷ with or without attention, the reaction vector readouts were passed to a multilayer perceptron (MLP) of depth = 2.⁵⁸ The final prediction is made as a single output vector with one entry for each label in the reaction dictionary, and the result is translated as described in Section 2.2.

3. RESULTS AND DISCUSSION

3.1. Model Performance. Our modeling pipeline was first tested on the Suzuki coupling data set, the largest of the four. Table 2 summarizes top-1 and top-3 categorical accuracies (eq 1) and AERs (eq 2) for the following models: GBMs with no trellising (BM-GBM), GBMs with trellising (CT-GBM), standard R-GCNs as reported by Schlichtkrull et al. (R-GCN),^{32,34} our AR-GCNs developed here (AR-GCN), and the dummy predictor as a baseline (dummy).

For this data set, GCN models significantly outperformed GBMs across categories for both top-1 and top-3 predictions. While GBMs actually gave negative top-1 AERs over baseline, these scores were dominated by the additive contribution; excluding this category, the BM- and CT-GBMs gave modest 10 and 9% AERs, respectively. Despite struggling with top-1 predictions, GBMs gave significant AERs for top-3, with BM-GBMs at 41% and CT-GBMs at 38%. The AR-GCNs gave the best accuracy of all models, providing 31 and 52% top-1 and top-3 AERs, respectively. AR-GCNs gave roughly 3% AER gain over the R-GCN in both top-1 and top-3 predictions, demonstrating the value of the added attention layer.

A few interesting categorical trends can be seen across model types. For instance, models provide the best error reduction ($\text{ER} = \frac{A_c^s - A_c^d}{1 - A_c^d}$, see eq 2) in the metal category, with the AR-GCN at 44 and 57% for top-1 and top-3, respectively. Similarly, models perform well in the base category, where the AR-GCN gave the best top-1 ER and BM-GBMs gave the best top-3 ER. Less consistent ERs between top-1 and top-3 predictions were obtained for the remaining three categories. For example, with solvents, the AR-GCN improved baseline by 23% in top-1 predictions, but 44% in top-3. Likewise, for AR-GCN ligand predictions, a 28% ER was obtained for top-1 versus a 56% gain in top-3. Finally, although the baseline additive accuracy is high as the majority of reactions are null in this category, the AR-GCN still gave a 23% top-1 ER and a 70% top-3 ER.

The trends and differences between top-1 and top-3 performance gains are reflective of the frequency distributions in each label category.³² These intuitively resemble long-tail or Pareto-type distributions,⁵⁹ with the bulk of the cumulative density contained in a small number of bins and the remaining bins supporting smaller frequencies. The distribution shapes are likely to influence the relative top-1 and top-3 AERs, where the highly skewed distributions could be more difficult to improve over baseline.

Having demonstrated the utility of our predictive framework, we turned to the remaining data sets to assess its scope. Modeling results for C–N, Negishi, and PKRs are detailed in Table 3 and Figure 4. Notable observations for each data set are discussed below.

3.1.1. C–N Coupling. Similar to the Suzuki results, the AR-GCN was the top performer for C–N couplings in almost all categories, and slightly higher AERs were observed overall. The AR-GCN afforded 36 and 55% top-1 and top-3 AERs,

Table 3. Prediction Accuracy for All Model Types on the C–N, Negishi, and PKR Data Sets

data set	top-k	category	dummy	BM-GBM	CT-GBM	R-GCN	AR-GCN
C–N	top-1	AER		−0.0413 ^a	−0.1593 ^b	0.3453	0.3604
		metal	0.2452	0.4825	0.4582	0.5989	0.6162
		ligand	0.5219	0.5538	0.5710	0.6981	0.7068
		base	0.2479	0.5028	0.5003	0.5932	0.6066
		solvent	0.3219	0.4582	0.4524	0.5647	0.5674
	top-3	additive	0.8904	0.7669	0.7031	0.8984	0.8997
		AER		0.3568	0.3131	0.5391	0.5471
		metal	0.6526	0.7928	0.7772	0.8479	0.8490
		ligand	0.6647	0.7933	0.7928	0.8605	0.8688
		base	0.6400	0.8008	0.7916	0.8452	0.8370
Negishi	top-1	solvent	0.5677	0.7370	0.7281	0.7973	0.7997
		additive	0.9156	0.9290	0.9184	0.9534	0.9559
		AER		0.3510	0.2773	0.4439	0.4565
		metal	0.2887	0.5444	0.5218	0.6555	0.6730
		ligand	0.7879	0.8174	0.7900	0.8724	0.8772
	top-3	temperature	0.3317	0.6656	0.6527	0.6188	0.6507
		solvent	0.6938	0.8562	0.8514	0.8868	0.8915
		additive	0.8309	0.8691	0.8401	0.8724	0.8644
		AER		0.5947	0.5199	0.6590	0.6833
		metal	0.5008	0.7771	0.7674	0.8086	0.8517
PKR	top-1	ligand	0.8549	0.9548	0.9321	0.9522	0.9553
		temperature	0.5885	0.9031	0.8772	0.8517	0.8708
		solvent	0.8788	0.9321	0.9402	0.9537	0.9537
		additive	0.9043	0.9548	0.9354	0.9761	0.9729
		AER		0.4396	0.4010	0.3973	0.4199
		metal	0.4302	0.7901	0.7786	0.7132	0.7057
		ligand	0.8792	0.9351	0.9237	0.9057	0.9094
		temperature	0.2830	0.5954	0.5649	0.6528	0.6642
		solvent	0.3321	0.6183	0.6260	0.6792	0.6981
	top-3	activator	0.6906	0.8244	0.8015	0.8415	0.8491
		CO (g)	0.7245	0.8855	0.8855	0.8717	0.8868
		additive	0.9057	0.9008	0.8893	0.8906	0.8491
		pressure	0.6528	0.8588	0.8702	0.8491	0.8491
		AER ^c		0.6987	0.6740	0.6844	0.7145
		metal	0.7132	0.9351	0.9313	0.9057	0.8906
		ligand	0.9019	0.9962	0.9924	0.9849	0.9962
		temperature	0.5962	0.8740	0.8321	0.8528	0.8604
		solvent	0.5925	0.8779	0.8550	0.8679	0.8981
		activator	0.8830	0.9466	0.9275	0.9774	0.9774
	top-3	CO (g)	1.0000	1.0000	1.0000	1.0000	1.0000
		additive	0.9321	0.9885	0.9885	0.9698	0.9736
		pressure	0.9623	0.9771	0.9847	0.9849	0.9849

^aAER excluding additive: 0.2302. ^bAER excluding additive: 0.2282. ^cExcludes CO(g). Bold numbers indicate the highest performing model in each dataset category.

respectively, again providing slight gains over R-GCNs at 35 and 54%. As above, GBMs struggled with this relatively large data set (36 519 reactions) due to difficulties with the additive category. Models again made strong improvements in the metal and base categories but also gave consistently strong gains for ligands⁴⁹ and solvents, especially for top-3 predictions. For example, the AR-GCN returned top-3 ERs of 57% for metals, 61% for ligands, 55% for bases, and 54% for solvents. Note that these ERs correspond to very high accuracies (A_c) of 85, 87, 84, and 80%, respectively.

3.1.2. Negishi Coupling. The highest AERs of all modeling experiments came with the Negishi data set. The AR-GCN again gave the strongest performance, with top-1 and top-3 AERs of 46 and 68%, respectively. However, the R-GCN and even GBM models gave the highest accuracies in some

categories. Interestingly, BM- and CT-GBMs performed significantly better than the GCNs for temperature predictions, though the strongest ER for most models came from the solvent category.

3.1.3. PKR. For the PKR data set, the smallest of the four, simple BM-GBMs gave the best top-1 AER at 44%, followed closely by the AR-GCN at 42%. Similarly for top-3 predictions, these models gave AERs of 70 and 71%, respectively. Compared to the other reactions, GCNs are perhaps more prone to overfitting this small data set,⁶⁰ making tree-based modeling more suitable. It is interesting to note that in general for PKRs, the GCN models were better at predicting physical parameters like temperatures, solvents, and CO(g) atmosphere, whereas GBMs gave better performance for reaction components such as metals, ligands, and additives.

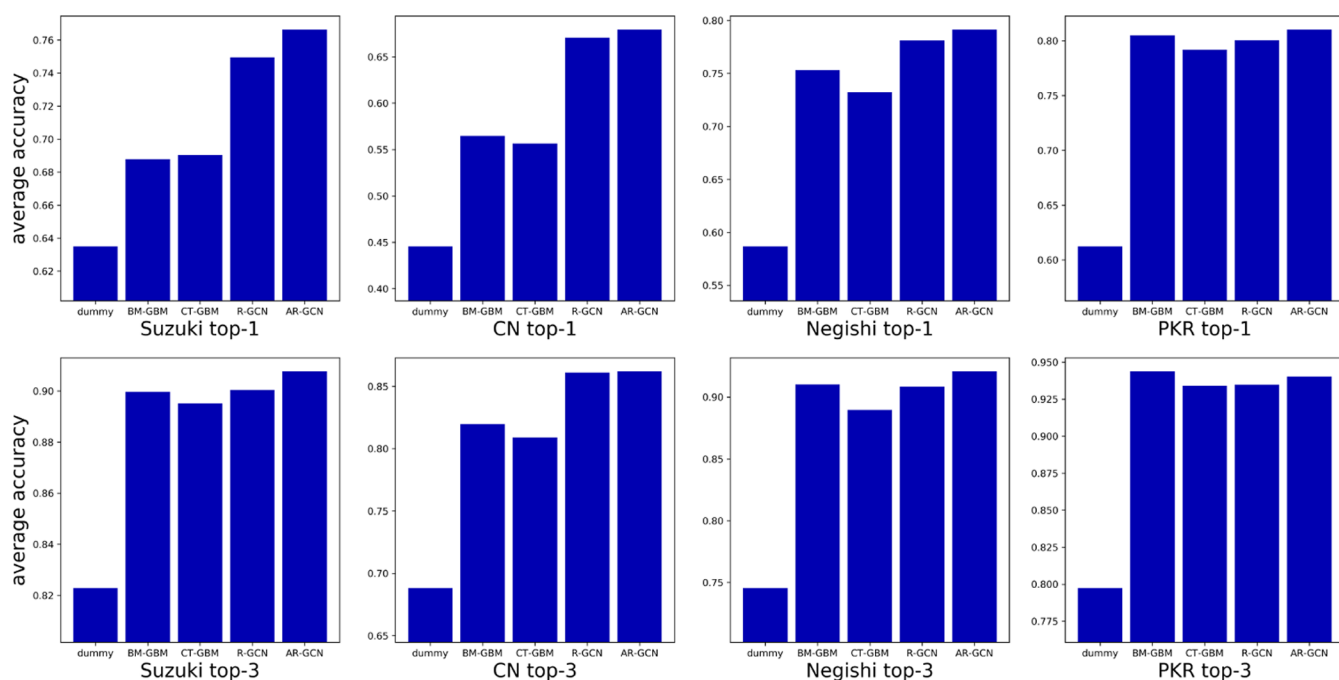


Figure 4. Average top-1 and top-3 categorical accuracies for each model across the four data sets.

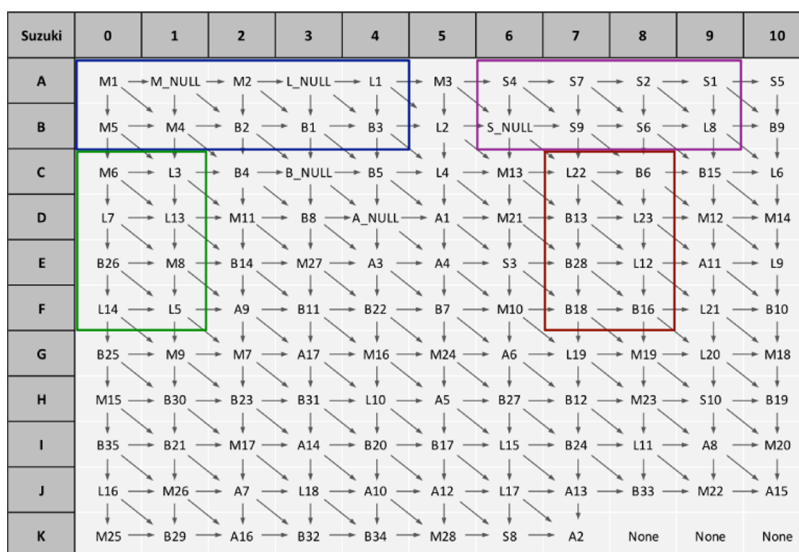


Figure 5. Optimized prediction trellis for the Suzuki data set.

3.2. Interpretability. **3.2.1. Tree Methods.** Given the results described above, we sought an understanding of the chemical features informing our predictions. Tree-based learning is often favored in this regard in that feature importances (FIs) can be directly extracted from models. We found that FIs for our BM-GBMs were roughly uniform across the SMILES regions of the encodings. The most informative physical descriptors from the Mordred vectors pertained to two classes: topological charge distributions⁶¹ correlated with local molecular dipoles and Moreau–Broto autocorrelations⁶² weighted by polarizability, ionization potential, and valence electrons (see the SI for detailed rankings). The latter class is particularly intriguing as they are calculated from molecular graphs in what has been described as atom-pair convolutions,⁶³ not unlike the GCN models used here.³⁴

An advantage to using CTs is the ability to extract their MI matrices and trellis structures for interpretation.⁵² The optimized trellis for the Suzuki CT-GBMs is included in Figure 5, where several chemically intuitive features and category blocks can be noted:

1. Block A0–B4 (blue): The result of M1 ($\text{Pd}(\text{PPh}_3)_4$) is used to predict three more metals: M2 ($\text{Pd}(\text{OAc})_2$), M4 ($\text{Pd}(\text{dppf})\text{Cl}_2\cdot\text{DCM}$), and M5 ($\text{Pd}(\text{PPh}_3)_2\text{Cl}_2$). Based on these metal complexes, the probability of using exogenous ligand (L_NULL) and L1 (PPh_3) is then predicted.
2. Block C0–F2 (green): The use of unligated M6 ($\text{Pd}_2(\text{dba})_3$) informs the predictions of ligands L3 (XPhos), L7 ($[(t\text{-Bu})_3\text{PH}]\text{BF}_4$), and L13 ($^{\text{Me}}\text{CgPPh}$). These in turn feed the model of unligated M8 ($\text{Pd}(\text{dba})_2$), which then informs L5 ($\text{P}(o\text{-tolyl})_3$).

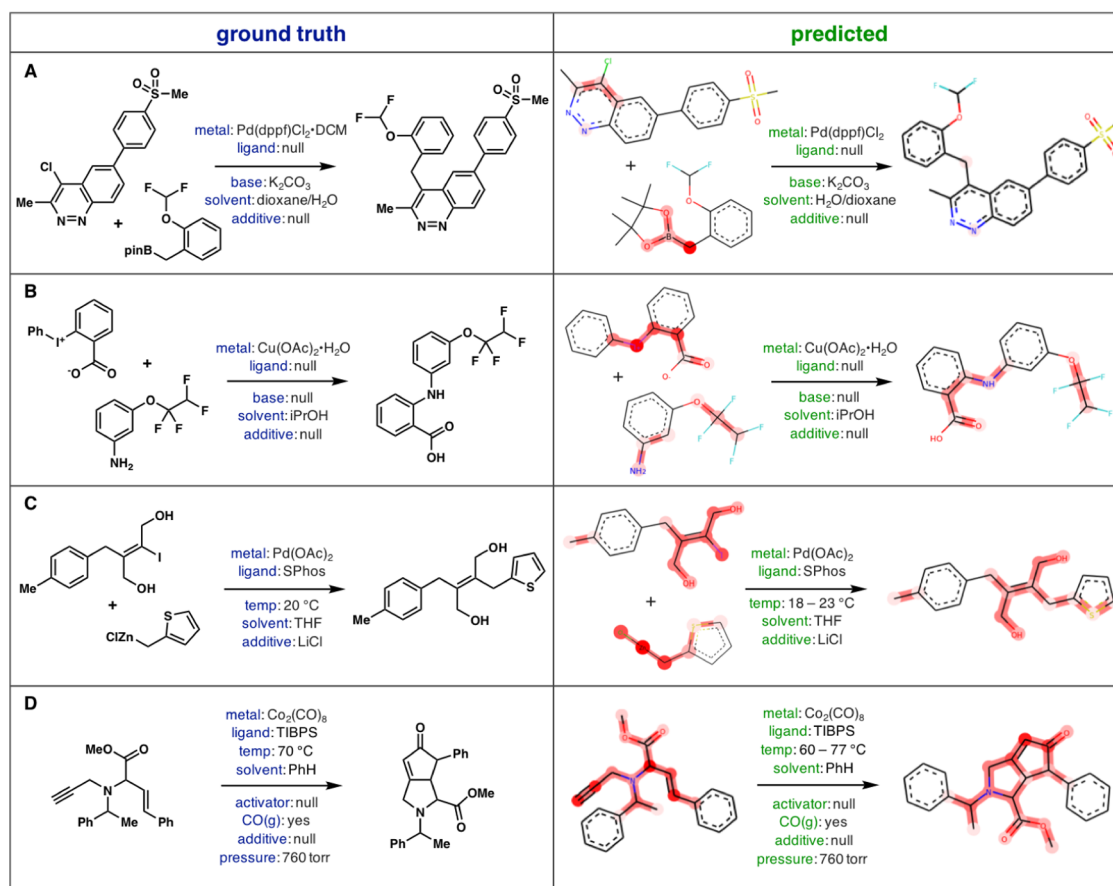


Figure 6. AR-GCN attention weight visualization and prediction examples from randomly chosen reactions in each data set. Darker highlights indicate higher attention.

- Block A6–B9 (purple): Several solvents are connected, where the predictions of S4 (1,4-dioxane) and S7 (PhMe) propagate through S9 (H₂O), S2 (EtOH), and S6 (MeCN). These additionally feed classifiers of S1 (THF) and S_NULL (neat).
- Block C7–F8 (red): Four different classes of bases are interwoven, including B6 (CsF) and B13 (KO^t-Bu). This informs the prediction of B28 (LiOH·H₂O), which then goes on to feed models of B18 (DIPEA) and B16 (NaO^t-Bu).

As a control experiment,⁶⁴ we withheld the propagated predictions from the CT-GBMs to test whether the MI was actually being used.⁶⁵ Indeed, model accuracy dropped off markedly, even below baseline in some categories. While this suggests that CT-GBMs do learn reagent correlations, the sharp performance loss may also indicate overfitting to this information.⁵² This is supported by the strong emphasis on the label-encoding region in the CT-GBM FIs. We hypothesize that reducing the dimensionality of the structure-encoding region could help balance model attention and avoid overfitting. Further studies are necessary to uncover the optimal molecule featurization in combination with CTs, though the results here suggest their promise in modeling structured reaction data.

3.2.2. Deep Learning Methods. For AR-GCNs, a valuable interpretability feature lies in the learned feature weights α'_i (eq 9). Intuitively, the weights represent the model's assignment of importance on an atom, as they rescale node features in the

final graph layer before inference. When extracted, the weights can be mapped back onto a molecule's atoms and displayed by color scale using RDKit (Figure 1D).⁶⁶ This gives a visual interpretation of the functional groups most heavily informing the predictions. Example visualizations from a random reaction in each data set and their AR-GCN predictions are included in Figure 6, and several additional random examples for each reaction type can be found in the SI.

In the Suzuki example (Figure 6A), the attention is dominated by the sp³ carbon bearing the Bpin group, with additional contributions from the bis-*o*-substituted heteroaryl-chloride and its cinnoline nitrogen, all of which could be reasonably expected to influence reactivity. It is interesting that weights on the *o*-difluoromethoxy group, the sulfone, and the majority of the product are suppressed, perhaps indicating that an alkyl nucleophile is sufficient to predict the required conditions. The AR-GCN predictions are correct in each category besides the metal, where the model erroneously identifies the metal source Pd(dppf)Cl₂ instead of its ground-truth DCM adduct Pd(dppf)Cl₂·DCM.

Conversely, the weights in the C–N coupling example are more evenly distributed (Figure 6B). Intuitively, the chemically active iodonium benzoate is given strong attention in the electrophile, as is the nucleophilic aniline nitrogen. Here, the *m*-tetrafluoroethoxy group is also weighted significantly and these groups are given similar attention in the product. All categories are predicted correctly in this example, though three of them are null.

The Negishi example (Figure 6C) is an interesting $C(sp^3)$ – $C(sp^2)$ coupling of a fully substituted alkenyl-iodide and thiophenyl-methylzinc chloride. Like with A, the strongest weights correspond to the sp^3 nucleophilic carbon, though similarly strong attention is distributed over the electrophilic alkene including the pendant alcohols. These weights are again reflected in the product, and all five condition categories are predicted correctly, including temperature and use of a LiCl additive.

Finally, an intramolecular PKR (Figure 6D) showed the most uniformly distributed attention of the four examples. Still, the strongest weights are given to the participating alkyne and alkene, with additional emphasis on the amino ester bridging group. Weights are similarly distributed in the product, though the strongest attention is intuitively assigned to the newly formed enone. Here, all eight categories are predicted correctly including the use of an ambient carbon monoxide atmosphere (CO(g) and pressure).

3.3. Yield Analysis. Having explored our models' chemical feature learning, we lastly investigated the effect of reaction yield, as it is a critical feature of synthesis data. Unsurprisingly, plotting the distribution of reaction yields in each data set showed a uniformly strong bias toward high-yielding reactions (Figure 7A). Given the skewness of the data in this regard, we hypothesized that models would perform best at predicting conditions for high-yielding reactions.

We divided the data set into quartiles by reaction yield and retrained the AR-GCN with each subset, subsequently testing in each region and on the full test set (Figure 7B). Intuitively, models trained in any yield range tended to give the highest accuracy when tested in the same range, occupying the confusion matrix diagonal in Figure 7B (top). To our surprise, however, the standard model trained on the full data set gave consistently high accuracies, regardless of the test set (bottom row).

Since the yield bins contain varying amounts of data, we re-split the data set, again ordered by yield but with equal subset sizes (Figure 7B, bottom). A similar trend was observed where the highest accuracies were found on the diagonal and bottom rows of the confusion matrix. Interestingly, the worst-performing model was that trained in the highest yield range and tested in the lowest. We recognize that making "inaccurate" predictions on low-yielding reactions offers an avenue for predictive reaction optimization and future studies will explore this objective.

4. CONCLUSIONS AND OUTLOOK

In summary, we present a multilabel classification approach to predicting experimental reaction conditions for organic synthesis. We successfully model four high-value reaction types using expert-crafted label dictionaries: Suzuki, C–N, and Negishi couplings, and Pauson–Khand reactions. We explore and optimize two model classes: gradient-boosting machines and graph convolutional networks. We find that GCN models perform very well in larger data sets, while GBMs show success for smaller data sets.

We report the first use of classifier trellises in molecular machine learning and find that they are able to incorporate label correlations in modeling. We introduce a novel reaction-level graph attention mechanism that provides significant accuracy gains when coupled with relational GCNs and construct a hybrid GCN architecture called attended relational GCNs or AR-GCNs. We further provide an analytical

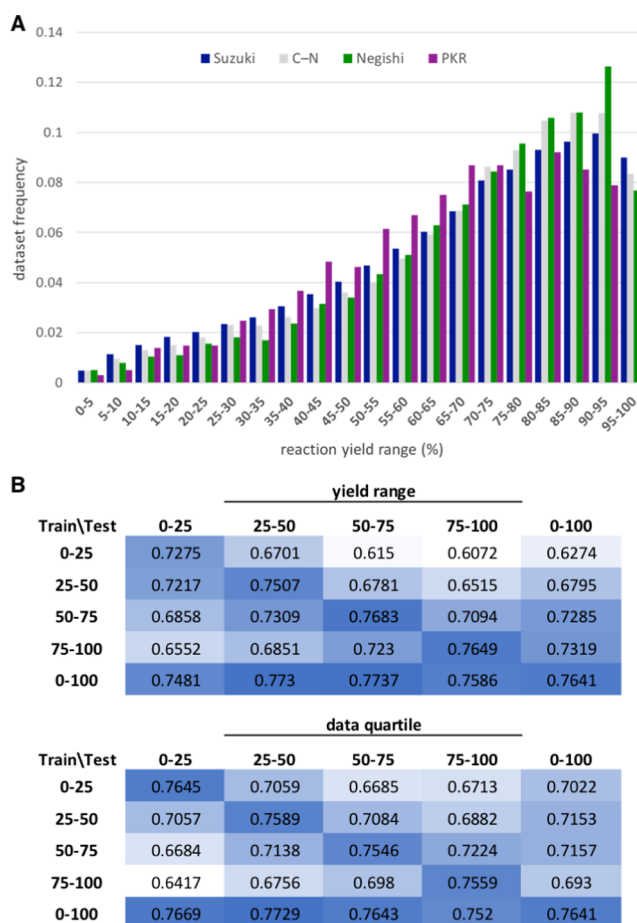


Figure 7. Performance dependence on reaction yields. (A) Distribution of reaction yields for the four data sets. (B) AR-GCN average top-1 A_c values for Suzuki predictions when trained and tested in different yield ranges (top) and data set quartiles arranged by yield (bottom).

framework for the chemical interpretation of our models, extracting the trellis structures and mutual information matrices of the CT-GBMs and visualizing the attention weights assigned in AR-GCN predictions.

Experimental studies are currently underway assessing the feasibility of model predictions on novel reactions. Additionally, efforts to apply our modeling framework to less-structured reaction types such as oxidations and reductions are ongoing. Future studies will address the interplay between structure representation and classifier chaining, as well as the extension of our reaction attention mechanism to other tasks. We expect the work herein to be very informative for future condition prediction studies, a highly valuable but underexplored learning task.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.0c01234>.

Data preparation and reaction dictionaries; computational details and hyperparameters; optimized classifier trellises, mutual information matrices, and reaction dictionaries for all data sets (PDF)

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Notes

The authors declare no competing financial interest.

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